

Functions and Function Notation



Function notation

- 1 Consider the function $f(x) = 2x^2 - 5x + 3$. Evaluate:

a $f(0)$	b $f(2)$	c $f(-1)$	d $f(\frac{1}{2})$
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 - 2 Consider the function $g(x) = \sqrt{x + 5}$. Evaluate:

a $g(4)$	b $g(-1)$	c $g(11)$	d $g(-5)$
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 - 3 Consider the function $h(x) = \frac{x-3}{x+2}$. Evaluate:

a $h(0)$	b $h(3)$	c $h(-1)$	d $h(5)$
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 - 4 If $f(x) = 3x - 7$ and $f(a) = 11$, find the value of a .
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Identifying functions

- 5 Determine whether each equation defines y as a function of x :

a $y = 2x + 1$	b $x = y^2$
c $y = x $	d $x^2 + y^2 = 9$
 - 6 Explain how the vertical line test is used to decide whether a graph represents a function.
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Combining functions

- 7 Let $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. Find:

a $(f + g)(x)$	b $(f - g)(x)$
c $(f \cdot g)(2)$	d $(f \circ g)(1)$
- 8 A tutoring center charges a one-time registration fee plus an hourly rate, modeled by $C(n) = 250 + 45n$ riyals for n hours of tutoring.

a Interpret the value $C(0)$ in context.	b Find $C(12)$.
c Solve $C(n) = 925$ and interpret your answer.	

- 9 Let $f(x) = \sqrt{x}$ and $g(x) = x - 4$. Find the domain of $(f \circ g)(x)$.
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Piecewise-defined functions

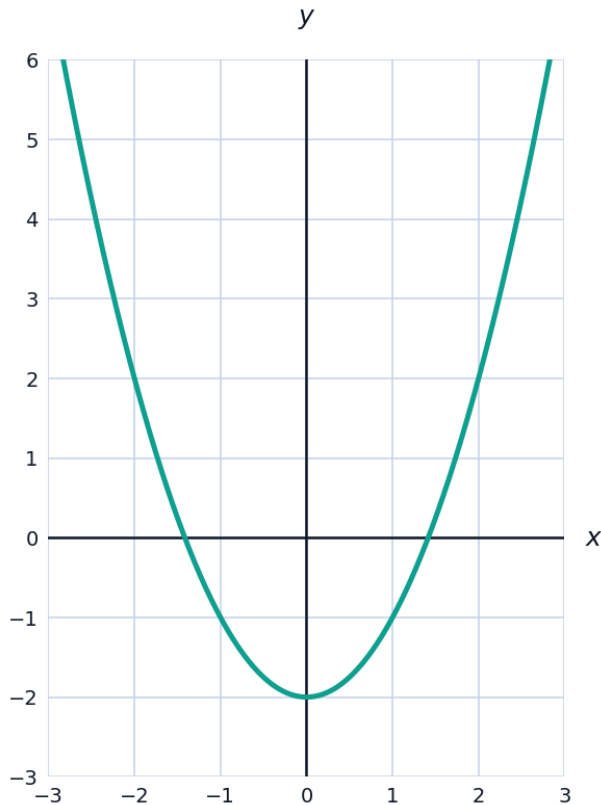
- 10 A function is defined by $f(x) = x + 4$ for $x < 1$, and $f(x) = 2x + 1$ for $x \geq 1$. Evaluate:
- a $f(-3)$ b $f(0)$ c $f(1)$ d $f(4)$
- 11 A phone plan charges a flat 60 riyals for up to 5 GB of data, then 15 riyals per additional GB beyond 5 GB.
- a Write a piecewise formula $C(x)$ for the monthly cost, where x is data used in GB.
- b Find $C(3)$ and $C(8)$.
- c Is C continuous at $x = 5$? Explain.
- 12 Sketch-reasoning: the graph of $f(x) = |x - 2|$ can be written as a piecewise function without absolute value bars. Write it as such, and state $f(-1)$ and $f(5)$.
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Even, odd, and one-to-one functions

- 13 Determine whether each function is even, odd, or neither:
- a $f(x) = x^4 - 3x^2$
- b $f(x) = x^3 - x$
- c $f(x) = x^2 + 2x$
- 14 Determine whether each function is one-to-one (passes the horizontal line test). If not, give a horizontal line that crosses the graph twice.
- a $f(x) = 2x + 5$ b $f(x) = x^2$
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Reading function graphs

- 15 The graph of $f(x) = x^2 - 2$ is shown. Use it to estimate $f(1)$, and state the interval(s) where f is increasing.



- 16 Using the graph of $f(x) = x^2 - 2$ above, estimate all x -values where $f(x) = 2$.

Average rate of change from function notation

- 17 For $f(x) = x^2 + 1$, find the average rate of change of f from $x = 1$ to $x = 3$.
- 18 A ball's height is modeled by $h(t) = -5t^2 + 20t$ (meters, seconds). Find the average rate of change of height from $t = 1$ to $t = 2$, and interpret the sign of your answer.

Difference quotients

- 19 For $f(x) = x^2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$.
- 20 For $f(x) = 3x + 2$, find and simplify the difference quotient $\frac{f(x+h) - f(x)}{h}$, and explain why the result makes sense for a linear function.
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Inverse functions (introduction)

- 21 Find the inverse of $f(x) = 3x - 4$, and verify by checking $f(f^{-1}(x)) = x$.
- 22 Explain why $f(x) = x^2$ (with domain all reals) does not have an inverse function, but $f(x) = x^2$ restricted to $x \geq 0$ does.